

Vitamin, Macromineral, and Trace Mineral Reference Range



Vitamin	Recommended Concentration Range	Biological Test Sample
Provitamin A (Carotenoids)	Blood – Carotenoids: Deficient: $<1.5 \mu\text{mol}\cdot\text{L}^{-1}$ Excess: $>4.65 \mu\text{mol}\cdot\text{L}^{-1}$	Blood
B ₁ (Thiamin)	Blood - Erythrocyte transketolase activity coefficient (ETKAC): Normal: ≈ 1.0 Deficient: >1.25 Urinary B1: Poor status: $<133 \text{ nmol}\cdot\text{d}^{-1}$ Low status: $133\text{--}333 \text{ nmol}\cdot\text{d}^{-1}$	Blood and Urine
B ₂ (Riboflavin)	Blood – erythrocyte glutathione reductase activity coefficient (EGRAC): Adequate: <1.2 Low: $1.2\text{--}1.4$ Deficient: >1.4 Urinary – Riboflavin gram per creatine (B2 $\text{g}\cdot\text{Cr}^{-1}$) Low status: $50\text{--}72 \text{ nmol}\cdot\text{g}^{-1}\cdot\text{Cr}^{-1}$ Poor status: $<50 \text{ nmol}\cdot\text{g}^{-1}\cdot\text{Cr}^{-1}$	Blood and Urine
B ₃ (Niacin)	Urinary Metabolites: Good status: $17.5\text{--}46.7 \mu\text{mol}\cdot\text{d}^{-1}$ Low status: $5.8\text{--}17.5 \mu\text{mol}\cdot\text{d}^{-1}$ Poor status: $<5.8 \mu\text{mol}\cdot\text{d}^{-1}$	Urine
B ₅ (Pantothenic Acid)	Urinary B5: Good status: $>11.8 \text{ mmol}\cdot\text{d}^{-1}$ Poor status: $<5 \text{ mmol}\cdot\text{d}^{-1}$	Urine
B ₆ (Pyridoxine)	Blood – Pyridoxal 5' – phosphate (PLP): Inadequate status: <20 to $30 \text{ nmol}\cdot\text{L}^{-1}$ Blood – Erythrocyte aspartate transaminase activity coefficient (EASTAC): Good status: <1.60 (may be variation) Blood – Erythrocyte alanine transaminase activity coefficient (EALTAC): Good status: <1.25 Urinary: 4-pyridoxic acid Adequate status: $>3.0 \mu\text{mol}\cdot\text{d}^{-1}$	Blood and Urine
B ₇ (Biotin)	Urinary B7: Good status $18\text{--}77 \text{ nmol}\cdot\text{d}^{-1}$ Urinary bisnorbiotin: Good status: $11\text{--}39 \text{ nmol}\cdot\text{d}^{-1}$ Urinary 3-hydroxyisovaleric acid: Good status: $77\text{--}195 \mu\text{mol}\cdot\text{d}^{-1}$	Urine (May not be practical measure for athletes)

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Vitamin	Recommended Concentration Range	Biological Test Sample
B₉ (Folic acid)	Blood – Poor status: Serum folate: <7 nmol·L ⁻¹ RBC folate: <305 nmol·L ⁻¹ Plasma homocysteine: >16 μmol·L ⁻¹	Blood
B₁₂ (Cobalamin)	Blood Analysis of B ₁₂ Good status: 148 to 185 pmol·L ⁻¹ Poor status: <148 pmol·L ⁻¹ Blood Serum – Methylmalonic acid (MMA) Poor status: >280 nmol·L ⁻¹ Plasma Homocysteine: Poor status: >16 μmol·L ⁻¹ Blood – Holotranscobalamin II: Good status: 19–50 pmol·L ⁻¹	Blood
Vitamin C (Ascorbic acid)	Blood Vitamin C: Deficient: <11.4 μmol·L ⁻¹ Low: 11.4–23 μmol·L ⁻¹ Adequate: >23 μmol·L ⁻¹	Blood
Choline*	Blood Good status: Plasma choline 7–20 μmol·L ⁻¹ Plasma phosphatidylcholine 1–1.5 mmol·L ⁻¹	Blood (Low accuracy)
Vitamin A (Retinol, retinal, and retinoic acid)	Blood Vitamin A: Deficient: <0.7 μmol·L ⁻¹ Severely deficient: <0.35 μmol·L ⁻¹ Excess: >3.5 μmol·L ⁻¹	Blood
Vitamin D (D₂ - ergocalciferol and D₃ - cholecalciferol) Provitamin (7-dehydrocholesterol)	Blood Vitamin D: Deficient: <50 nmol·L ⁻¹ Insufficient: <75 nmol·L ⁻¹ Sufficient: >75 nmol·L ⁻¹ Optimal: >40 nmol·L ⁻¹	Blood
Vitamin E (Tocopherols and tocotrienols)	Blood Vitamin E: Deficient: <11.6 μmol·L ⁻¹ Adequate: 12–42 μmol·L ⁻¹ Excess: >464 μmol·L ⁻¹	Blood
Vitamin K (Phylloquinones and menaquinones)	Blood Clotting: Adequate: Plasma prothrombin time 10.5–12.5 sec Inadequate: Delayed clotting time	Blood

Adapted from Larson-Meyer et al. (2018).

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Macro or Trace Mineral	Recommended Concentration Range	Biological Test
Calcium (Ca)	No current reference range	—
Phosphorus (P)	Blood P: Serum P: 0.87–1.45 mmol·L ⁻¹	Blood (Low accuracy)
Potassium (K ⁺)	Blood K: 25–125 mmol·d ⁻¹	Blood
Sodium (Na ⁺)	Urinary Na: 40–220 mmol·d ⁻¹	Urine
Magnesium (Mg)	Blood Mg: Good status: Serum Mg: 0.74–1.07 mmol·L ⁻¹ Urinary Mg: 3–5 mmol·d ⁻¹	Blood and Urine (Mg test lack accuracy)
Iron (Fe)	Please refer to Table: Practical Blood Measures of Iron Status	Blood
Zinc (Zn)	Blood Zn: Good status: Serum Zn: 10.7–18.4 μmol·L ⁻¹ Urinary Zn: 2.3–18.4 μmol·d ⁻¹ Poor status: Serum Zn: <10.7 μmol·L ⁻¹	Blood (No current reliable test for Zn)
Selenium (Se)	Blood (Se): Reference range: 0.58–1.82 μmol·L ⁻¹	Blood (Low accuracy and no set ranges)
Chromium (Cr)	Blood Cr: Good status: Reference range: 1.9–38.4 nmol·d ⁻¹	Urine (Low accuracy)
Iodine (I)	Urinary I: Mild deficiency: 4–8 μmol·d ⁻¹ Moderate deficiency: 2–3.9 μmol·d ⁻¹ Severe deficiency: <2 μmol·d ⁻¹	Urine (24-hour urine collection may not be practical)

Adapted from Larson-Meyer et al. (2018).